Project Design Phase-I
Solution Architecture

Date	25 October 2026
Team ID	EN25CS3030037
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	The four project datasets: athletes.csv (8,000 athlete records with 26 fields), medal_results.csv (910 country results per Games with 19 fields), games_summary.csv (18 Games with 10 fields) and alltime_country_rankings.csv (60 countries with 13 fields). Together they cover 60 countries, 50 sports and the Games from 1992 to 2026.	CSV files
Data Connection	Connects Tableau to the CSV files and lets the data be refreshed when a file changes.	Tableau Data Connections (Text File)
Data Preparation	Checks data types, nulls and duplicates, relates the files on noc_code and year, assigns geographic roles to the country field and creates a Year field.	Tableau Data Source tab
Data Layer	Stores the prepared data model and extract that every worksheet and dashboard reads from.	Tableau data model / extract
Analytics / BI Layer	Calculates KPIs and measures such as total medals, medals per athlete, average medal efficiency, sports investment against medals and nations represented, and applies filters, aggregations and the KPI Selector parameter.	Tableau calculated fields and parameters
Visualization / Reporting	Presents the insights as charts in two interactive dashboards and a story, with filters and tooltips.	Tableau Desktop (Dashboard 1, Dashboard 2 and Story)
Web Front-End	Publishes the dashboards and story and shows them inside a small website.	Tableau Public, Flask (Python), HTML / CSS
Business Users	Sports strategists, investment analysts and journalists explore the dashboards to compare countries, sports and regions and to make decisions.	Tableau dashboards (Tableau Public / browser)